Lab-Assignment 6.4

NAME: P.RANADHEER REDDY

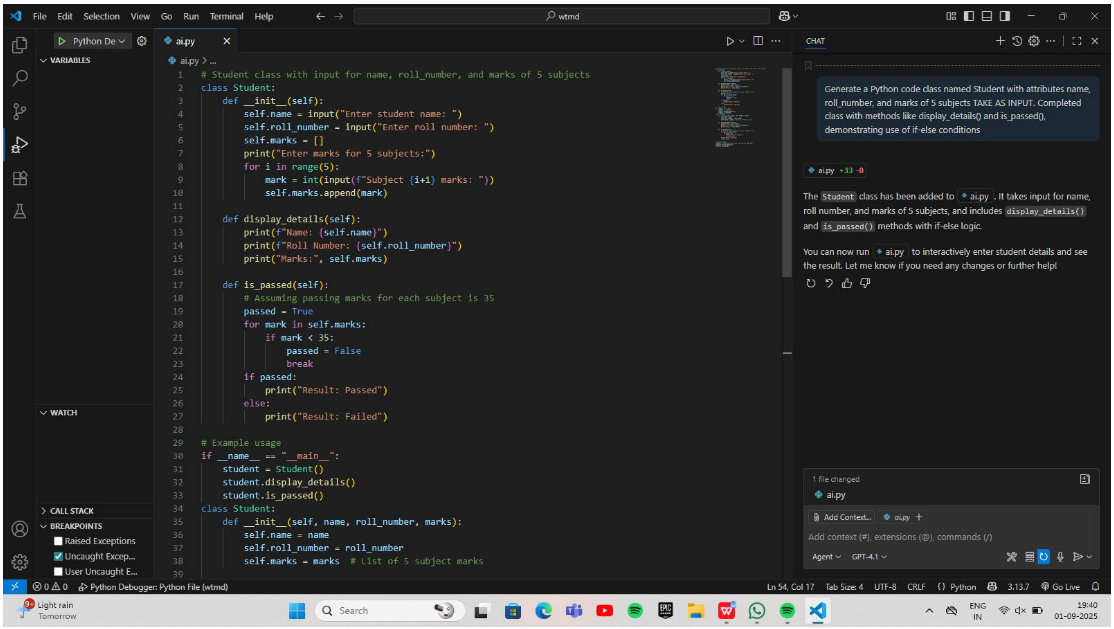
HT.NO: 2403A54098

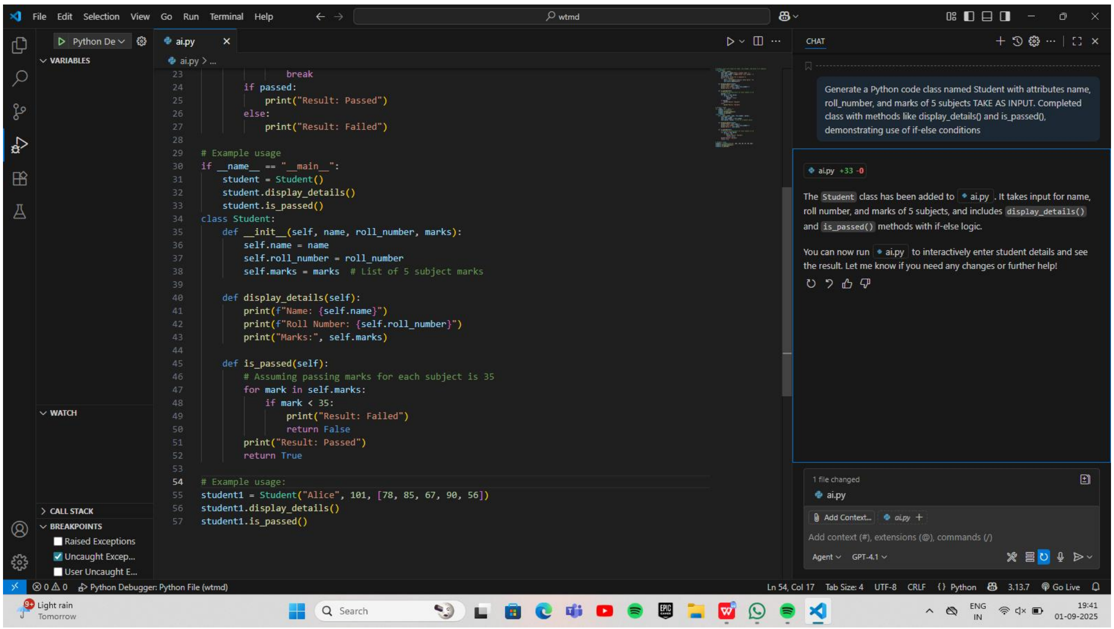
BATCH: 18

Task-1:

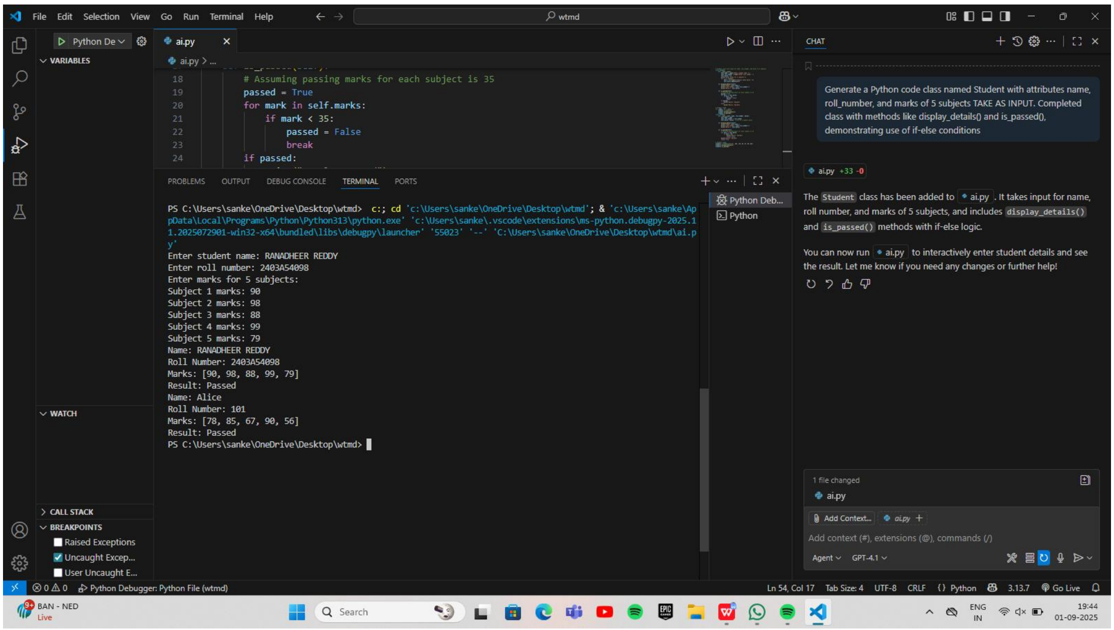
PROMPT: Generate a Python code class named Student with attributes name, roll_number, and marks of 5 subjects . Completed class with methods like display_details() and is_passed(), demonstrating use of if-else conditions.

CODE:





OUTPUT:

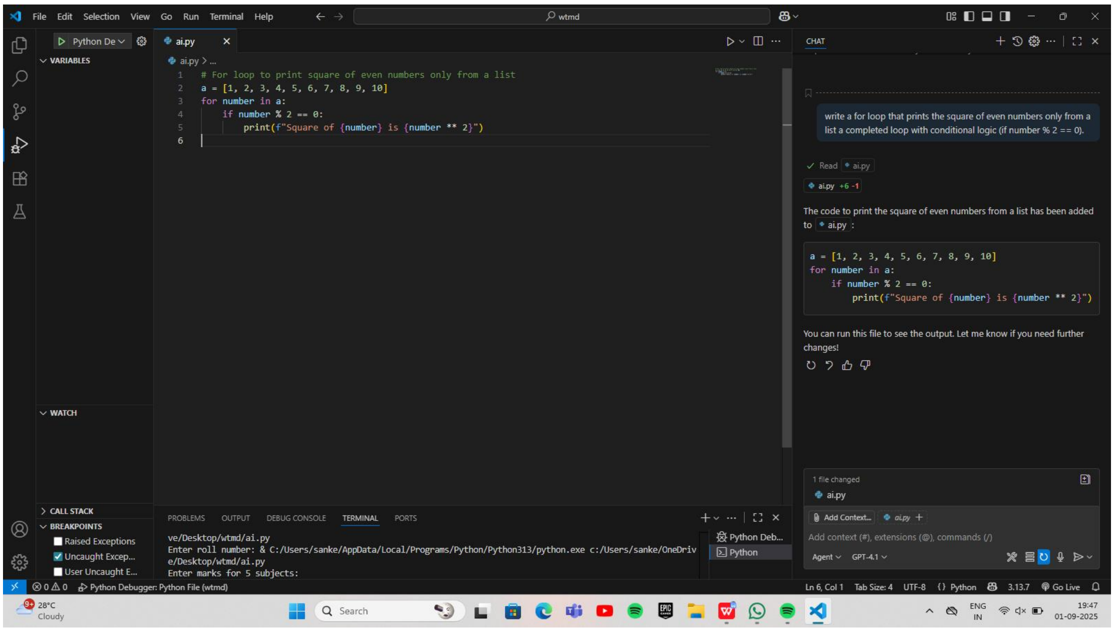


Task-2:

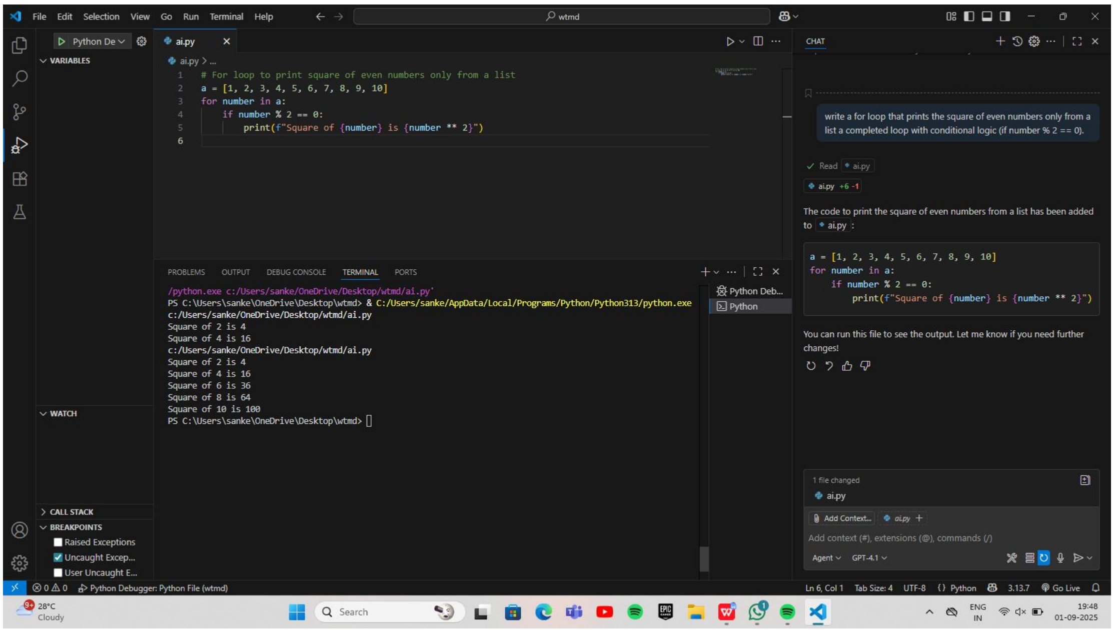
PROMPT:-

write a for loop that prints the square of even numbers only from a list a completed loop with conditional logic (if number % 2 == 0).

CODE:



OUTPUT:

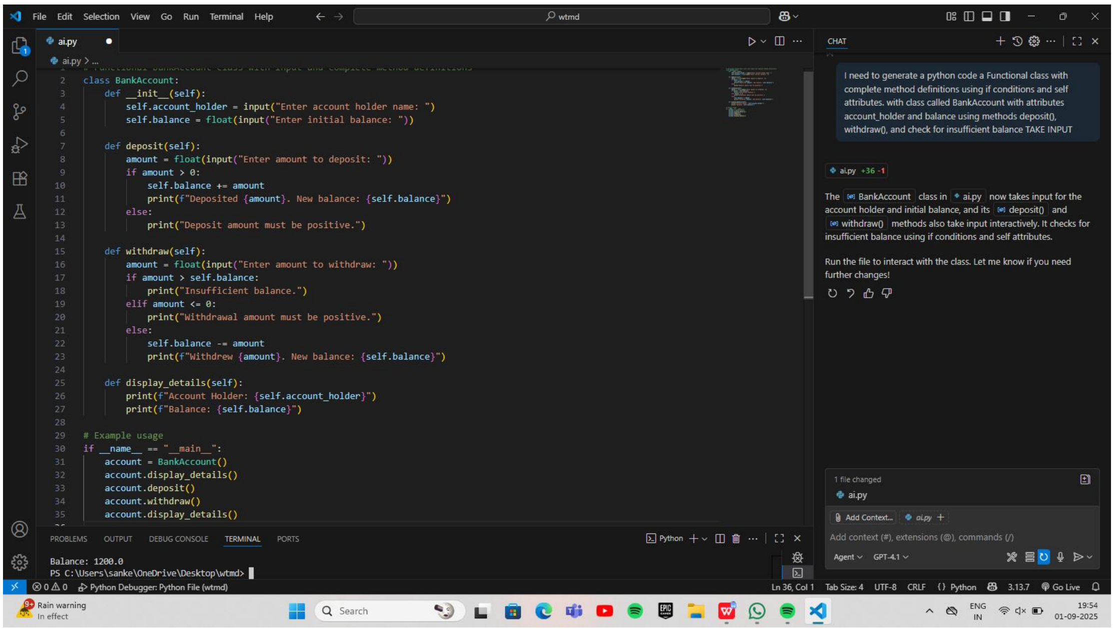


Task-3:

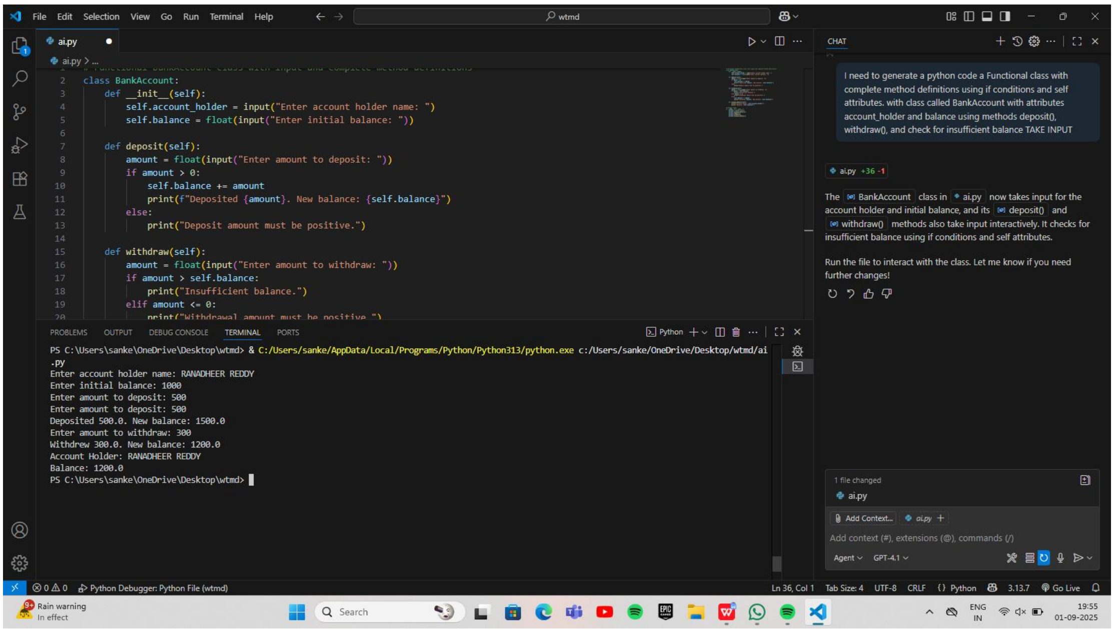
PROMPT:

I need to generate a python code a Functional class with complete method definitions using if conditions and self attributes. with class called BankAccount with attributes account_holder and balance using methods deposit(), withdraw(), and check for insufficient balance

CODE :-



OUTPUT:

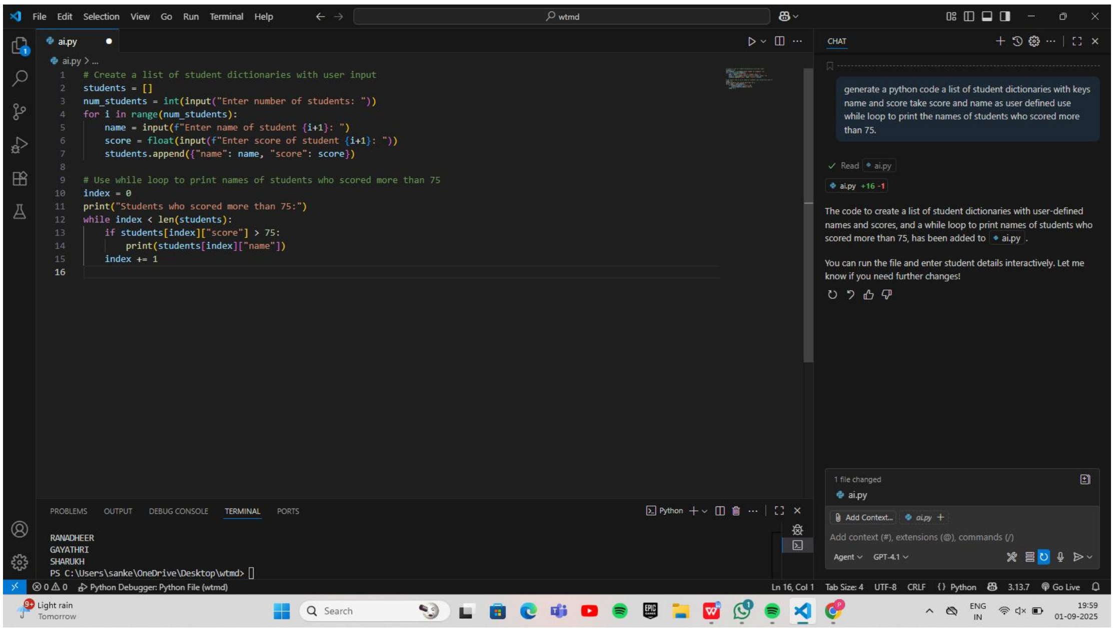


Task4:

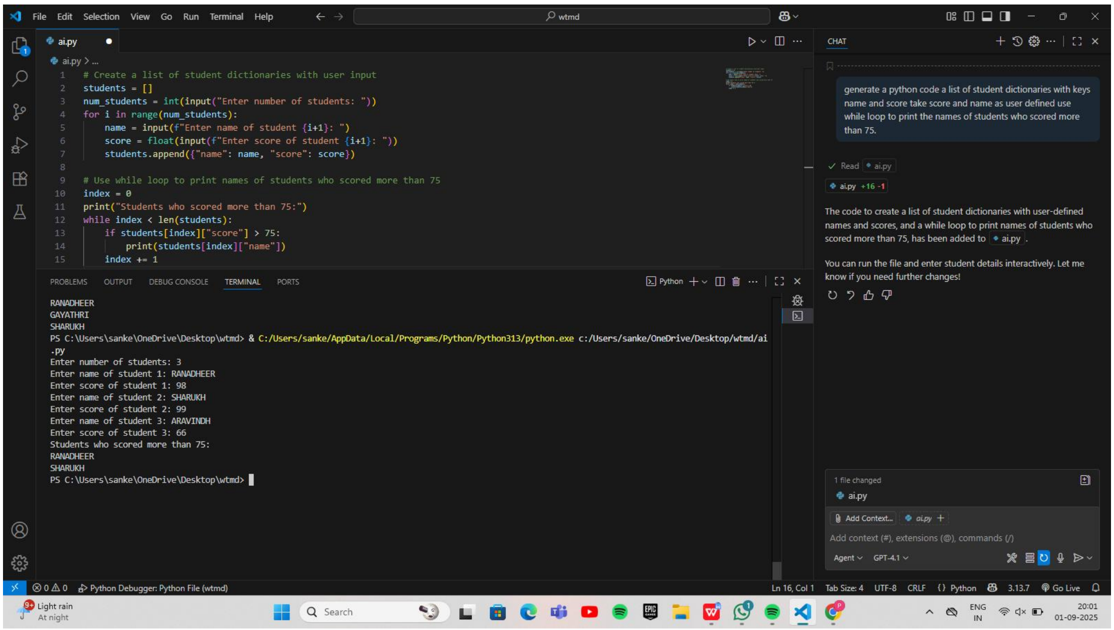
PROMPT:

generate a python code a list of student dictionaries with keys name and score take score and name as user defined use while loop to print the names of students who scored more than 75.

Code:



Output:

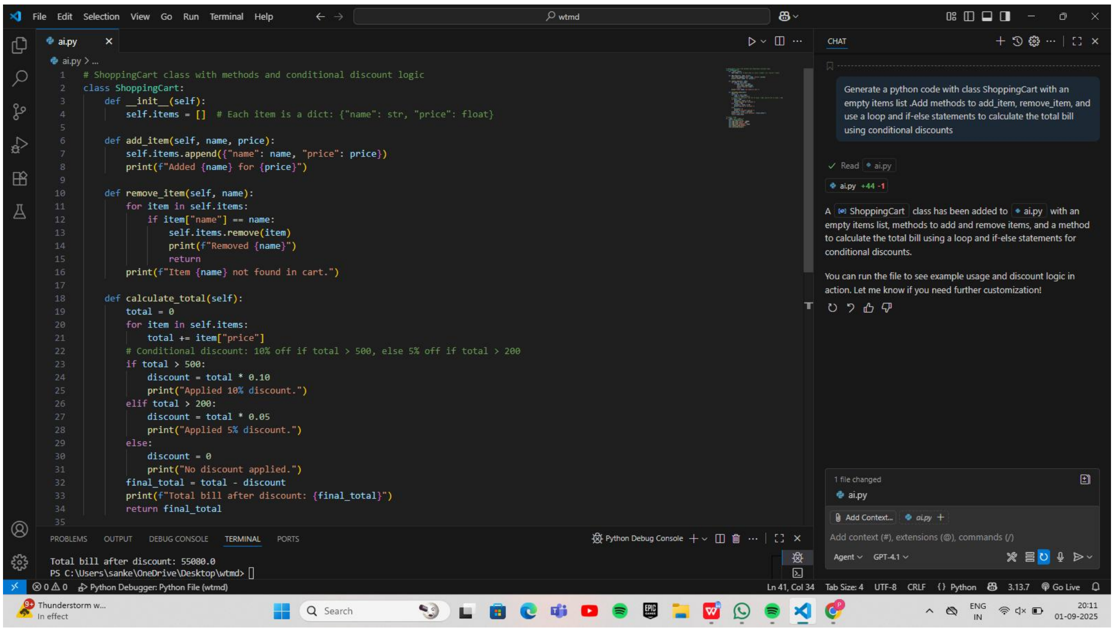


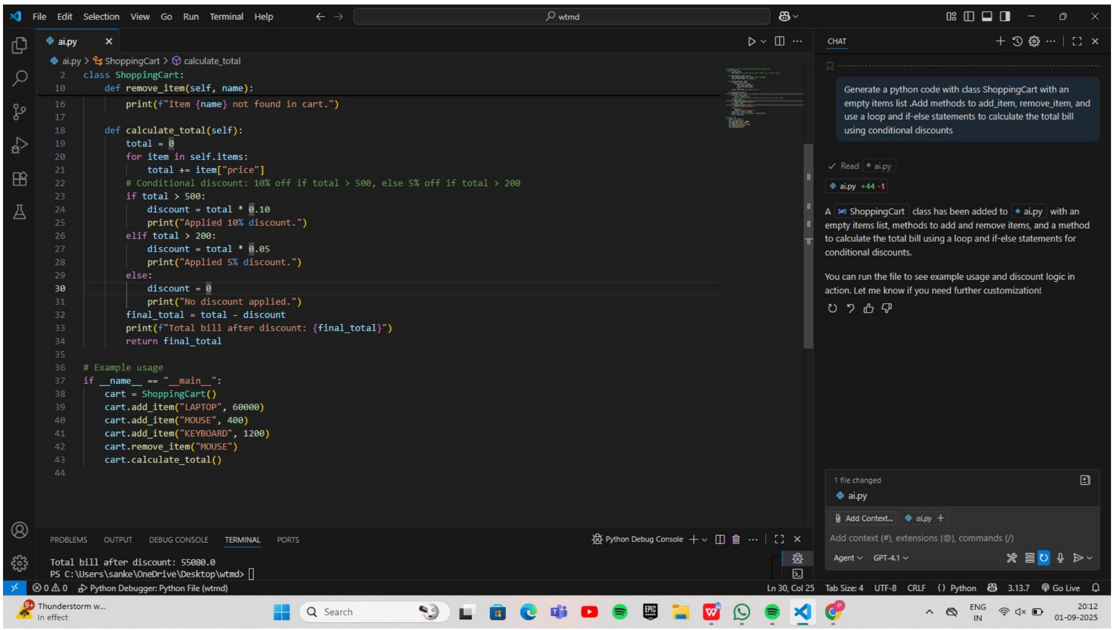
Task5:

Prompt:

Generate a python code with class ShoppingCart with an empty items list .Add methods to add_item, remove_item, and use a loop and if-else statements to calculate the total bill using conditional discounts

Code:


ai.py
# ShoppingCart class with methods and conditional discount logic
class ShoppingCart:
def __init__(self):
self.items = [] # Each item is a dict: {"name": str, "price": float}
def add_item(self, name, price):
self.items.append({"name": name, "price": price})
print(f"Added {name} for {price}")
def remove_item(self, name):
for item in self.items:
if item["name"] == name:
self.items.remove(item)
print(f"Removed {name}")
return
print(f"Item {name} not found in cart.")
def calculate_total(self):
total = 0
for item in self.items:
total += item["price"]
# Conditional discount: 10% off if total > 500, else 5% off if total > 200
if total > 500:
discount = total * 0.10
print("Applied 10% discount.")
elif total > 200:
discount = total * 0.05
print("Applied 5% discount.")
else:
discount = 0
print("No discount applied.")
final_total = total - discount
print(f"Total bill after discount: {final_total}")
return final_total
Total bill after discount: 55080.0
PS C:\Users\sanke\OneDrive\Desktop\wtmd>
CHAT
Generate a python code with class ShoppingCart with an empty items list .Add methods to add_item, remove_item, and use a loop and if-else statements to calculate the total bill using conditional discounts
A ShoppingCart class has been added to ai.py with an empty items list, methods to add and remove items, and a method to calculate the total bill using a loop and if-else statements for conditional discounts.
You can run the file to see example usage and discount logic in action. Let me know if you need further customization!


ai.py > ShoppingCart > calculate_total
class ShoppingCart:
def remove_item(self, name):
print(f"Item {name} not found in cart.")
def calculate_total(self):
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for item in self.items:
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discount = 0
print("No discount applied.")
final_total = total - discount
print(f"Total bill after discount: {final_total}")
return final_total
# Example usage
if __name__ == "__main__":
cart = ShoppingCart()
cart.add_item("LAPTOP", 60000)
cart.add_item("MOUSE", 400)
cart.add_item("KEYBOARD", 1200)
cart.remove_item("MOUSE")
cart.calculate_total()
Total bill after discount: 55080.0
PS C:\Users\sanke\OneDrive\Desktop\wtmd>
CHAT
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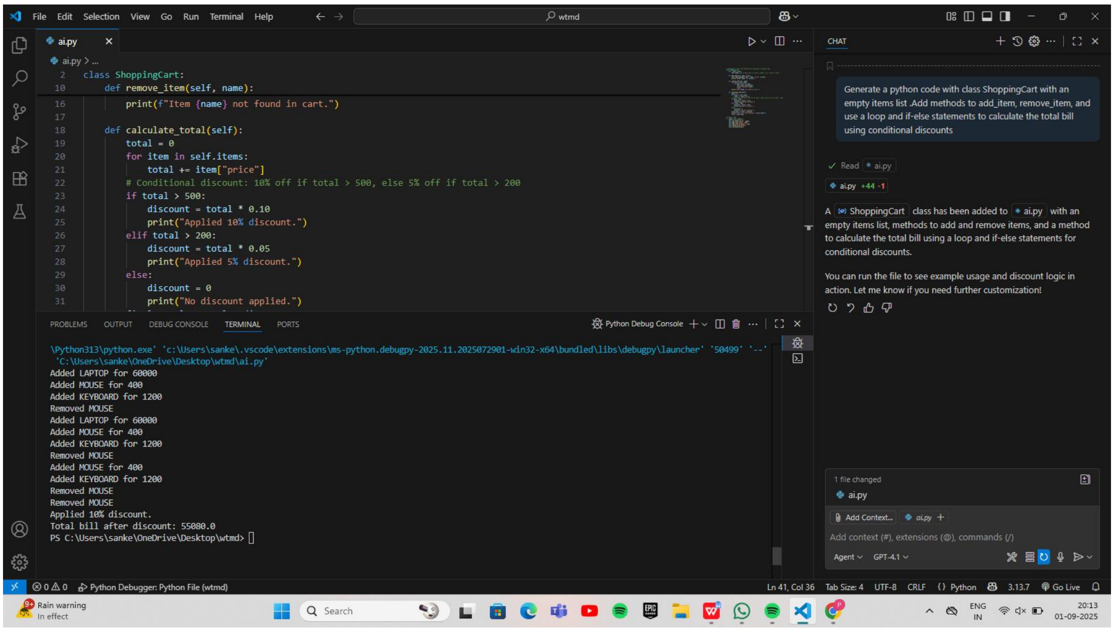
Output:



File Edit Selection View Go Run Terminal Help
wtmd
ai.py
ai.py > ...
class ShoppingCart:
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python Debug Console
\Python313\python.exe' 'c:\Users\sanke\.vscode\extensions\ms-python.debugpy-2025.11.2025072901-win32-x64\bundled\libs\debugpy\launcher' '50499' '--' 'C:\Users\sanke\OneDrive\Desktop\wtmd\ai.py'
Added LAPTOP for 60000
Added MOUSE for 400
Added KEYBOARD for 1200
Removed MOUSE
Added LAPTOP for 60000
Added MOUSE for 400
Added KEYBOARD for 1200
Removed MOUSE
Added MOUSE for 400
Added KEYBOARD for 1200
Removed MOUSE
Removed MOUSE
Applied 10% discount.
Total bill after discount: 55080.0
PS C:\Users\sanke\OneDrive\Desktop\wtmd>
CHAT
Generate a python code with class ShoppingCart with an empty items list .Add methods to add_item, remove_item, and use a loop and if-else statements to calculate the total bill using conditional discounts
Read ai.py
ai.py +44 -1
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You can run the file to see example usage and discount logic in action. Let me know if you need further customization!
1 file changed
ai.py
Add Context... ai.py
Add context (#), extensions (@), commands (/)
Agent GPT-4.1
Python Debugger: Python File (wtmd)
Ln 41, Col 36 Tab Size: 4 UTF-8 CRLF Python 3.13.7 Go Live
Rain warning
In effect
Search
ENG
IN
20:13
01-09-2025